ARTICLE

PRISM: an auditable pipeline for post-treatment glioma surveillance, evaluated branch by branch

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ABSTRACT

PRISM predicts documented progression within 120 days of a glioma surveillance MRI and returns an audit record and a clinician report per scan. Its first version scored AUC 0.80 on one MU-Glioma-Post test split but averaged 0.72 over 500 similar splits; that split was easy because of its mix of scans before radiotherapy. Under repeated cross-validation, clinical and treatment-course variables reached 0.70 in grade 4, or 0.68 after removing leaks found by its own audit. Radiomics, frozen and fine-tuned deep encoders, resegmentation, a cleaner label and diffusion showed no measurable gain, and a locked model scored 0.61 externally. The phase of care drove discrimination: in surveillance after radiotherapy, the intended use, AUC was 0.60. Reports written from the audit record by a local language model passed a faithfulness check for all 249 grade 4 scans and contained every prespecified content item in 247; clinicians have not yet assessed them.

KEYWORDS

glioma surveillance; prediction model evaluation; auditable decision support

1. Introduction

Most patients with high-grade glioma progress after surgery and chemoradiotherapy, and they are followed with MRI every few months. At each surveillance scan the team asks how likely it is that progression will be documented before the next decision point. Two things make this hard. Enhancement can increase after radiotherapy (RT) without tumour growth. This pseudoprogression is most common in the first 12 weeks after chemoradiotherapy and more frequent with MGMT promoter methylation (Brandes et al. 2008; Taal et al. 2008), which is why the RANO criteria ask for confirmation of progression in that window (Wen et al. 2010, 2023). And in routine data, progression is a documentation event. It depends on imaging, on symptoms and on when the next scan happens to be scheduled. A model for this endpoint is a prognostic aid for deciding whether to bring the next scan forward or act on an equivocal finding. For that use, a calibrated probability, an honest interval and a way to check what the model knew matter as much as a high AUC.

Most imaging work addresses a neighbouring question: telling pseudoprogression from true

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progression at a lesion that is already suspicious. There, machine learning models pool to an AUC of about 0.77 (Booth et al. 2022), and diffusion and perfusion measures reach 0.90 to 0.93 (Taylor et al. 2022). Forward prediction from every surveillance scan has had less attention. On the public MU-Glioma-Post dataset (Mahmoud et al. 2025), a calibrated radiomics model reported AUC 0.80 on one held-out set of patients (Christodoulou et al. 2026).

Three problems can make such numbers mislead. Single train–test splits are unstable in small radiomics cohorts: in one brain tumour dataset, one random split tested at 0.667 and another at 0.911 (An et al. 2021). Performance drops on external data, by 0.10 or more for about a quarter of externally validated deep learning algorithms (Yu et al. 2022). And clinical comparators are rarely reported, although where they are, they hold their own (Leone et al. 2023; Gomaa et al. 2025; Abler et al. 2025). A fourth problem is specific to surveillance. A cohort mixes scans taken before RT, during it, in the 12 weeks after it and in later follow-up, and event rates differ sharply between these phases of care. Time since the end of RT was an input to both of the best documented clinical comparators (Gomaa et al. 2025; Leone et al. 2023). A model that only recognises the phase can rank scans well whenever the test set holds many easy phases, and treatment fields recorded once per patient can leak the future into a scan taken before treatment started (Kapoor and Narayanan 2023).

We built PRISM (Post-treatment Response Intelligence and Surveillance Monitor) for this task. From what is known on the scan day, it estimates the risk of documented progression within 120 days, records how the estimate was made, and writes a short report that can be checked against that record. Its first evaluation, on one split of 30 test patients, gave an AUC of 0.80. We tested that number before relying on it, and each of the problems above applied. This paper describes PRISM and what a rigorous evaluation showed: why the single split misled, that the phase of care rather than MRI carries the signal, and how the audit record and the report make a modest model transparent and checkable.

2. Methods

2.1. PRISM

PRISM takes one MRI (T1, T1c, T2 and FLAIR with a four-compartment tumour mask) and the patient's clinical tables and returns three things for each scan: a risk, an audit record and a report (Figure 1). Every input is read as it stood on the scan day.

The risk. A ridge logistic model combines the inputs available on the scan day: clinical and molecular data, the treatment course and, optionally, imaging. The reference configuration is the *core model* of clinical and treatment-course variables (Section 2.3). Three imaging branches can be added: imaging measurements (volumes, morphology and change), radiomics, and a SwinUNETR image encoder, frozen or fine-tuned. Any branch can be switched off, which is how we ablate it.

The audit record. An audit layer wraps the model, locks it with a model card (code version, hashes of every input table, each input's definition, source and as-of rule, intended and excluded uses) and writes one record per scan. The record holds each input with its source and the day it became known; the phase of care; the risk with a 95% interval from 200 bootstrap refits; the exact contribution of each input group to the log odds; a split conformal prediction set at $\alpha = 0.2$; and referral flags with reasons. A scan is flagged if it falls in the 12 weeks after RT, if an input lies

outside the training range or a molecular marker is unknown, or if its conformal set is {0, 1}.

The report. A local language model (Qwen3.8-27B at a pinned revision, greedy decoding, run offline on our own GPUs) writes the report from one scan's audit record and nothing else, in the findings and impression form of radiology reports. Every number must be copied from a record field and every sentence must cite the fields it uses. Beyond the risk, the report states what a reader needs next to the number: how the risk compares with the model's average, the risk at the previous scan, the phase of care and why it matters, the main drivers, every active flag with its reason, what the model did not see (the images themselves), and any input not known on the scan day. Diagnoses, RANO categories, management advice and unhedged certainty are forbidden. A deterministic checker scores every sentence: cited fields must exist, every number must match a cited value, every active flag must be mentioned, no inactive flag may be described as raised, and no forbidden pattern may appear.

PRISM's first version used a calibrated radiomics model, an untrained image encoder fused with it, and a report from fixed RANO rules and a commercial language model. Its encoder carried no signal and its fusion did not beat the radiomics input (Supplementary S7); the components above replace them.

2.2. Data, outcome and intended use

MU-Glioma-Post (Mahmoud et al. 2025) was the development cohort: MRI after surgery from a single centre, with four-compartment masks (enhancing tumour, ET; non-enhancing core; surrounding FLAIR signal; resection cavity) and a clinical workbook giving the day of each scan, surgery, RT, chemotherapy, first progression and death, and age, sex, grade, IDH and MGMT. LUMIERE (Suter et al. 2022) was the external cohort: glioblastoma follow-up scans with an expert RANO rating, automatic segmentations, IDH and MGMT, but no treatment dates. CFB-GBM (Moreau et al. 2025) supported a diffusion analysis: 264 patients imaged at RT planning and twice afterwards, with ADC and volumetric RANO 2.0 labels; we used only its released tables.

An index scan is an MRI before first documented progression and before later lines of treatment, with all four sequences, a mask and a known date. Each patient contributes every eligible scan, and every analysis keeps a patient's scans together. Grade 4 glioma was the primary population, fixed in advance (Table 1; study design in Figure 2). The outcome is documented first progression within 120 days of the scan. A scan counts as negative only if follow-up without progression reaches 120 days; otherwise it is censored, and death without progression censors at the day of death. Discrimination is the AUC at 120 days weighted by the inverse probability of censoring (IPCW AUC; Uno et al. 2007; Blanche et al. 2013). In LUMIERE, progression is the first expert rating of progressive disease.

PRISM's intended use is surveillance after treatment has begun: scans taken on or after the documented start of RT, a condition known on the scan day. Scans before RT are scored but flagged as outside the intended use (Section 3.3).

Reporting follows TRIPOD+AI (Collins et al. 2024) and CLEAR (Kocak et al. 2023 [CHECK]; Supplementary S17). For each analysis called prespecified, the plan and its decision rule were written down before any of its model outputs were seen, and for the later analyses committed to the public project repository first; the rules and answers are collected in Table 4 (Supplementary S1). None was registered with an external registry.

2.3. Predictors

All predictors are defined as of the scan day (Supplementary S4). Clinical: age, sex, IDH and MGMT. Timing: days since diagnosis and since RT ended, a flag for the 12 weeks after RT and its interaction with MGMT methylation, and visit number. Treatment: whether RT and chemotherapy had started and whether RT was complete, the prescription, and the days of surgery and RT start. A treatment fact enters only once its documented start is on or before the scan day. These variables form the core model. A compact alternative replaces timing and treatment with four phases of care: before RT, during RT, within 12 weeks after RT, and later. Imaging measurements are compartment volumes, morphology and change from earlier scans only. Radiomics cover all four sequences (Supplementary S5). Deep features are pooled from SwinUNETR encoders with BraTS 2021 or BrainSegFounder weights (Cox et al. 2024 [CHECK]), with untrained weights as a control (Supplementary S6). Unknown values are imputed within each training fold.

2.4. Evaluation

One split, then many. The first evaluation used one split from PRISM's own split search: for a given seed, 20,000 random draws of 30 test patients, keeping the draw closest to 96 test scans and 53 positives. With the final cohort it holds 126 training and 29 test patients (82 test scans). We rescored it with the final labels and reran the same search for seeds 0 to 499, refitting the models on each split, to see where the first split ranks and how each split's AUC relates to its mix of phases (Supplementary S8).

Repeated cross-validation. All main estimates use one shared set of 5 × 5 repeated folds, grouped by patient and stratified by outcome and grade. Every fitted step (imputation, clipping, scaling, feature selection, stacking and hyperparameters) is fitted inside the training fold (Varma and Simon 2006; Cawley and Talbot 2010). The estimate is the mean over repeats of the pooled held-out IPCW AUC; 95% CIs and paired differences come from 1,000 bootstrap draws over patients, with all models scored on the same draws.

What carries the signal. Each branch is added to, or removed from, the core model. Radiomics enters through 70 configurations and frozen deep features through 32; one configuration is chosen inside each outer training fold by inner cross-validation (nested selection). To test the phase of care, we built bases without imaging (R0 clinical; R1 plus timing; R2 clinical plus phase; R3 the core model), added imaging blocks to each, and repeated the analysis within the surveillance cohort, the scans taken after RT was complete (grade 4: 167 scans, 84 patients). Because censoring turned out to be concentrated before RT, we re-estimated the censoring weights separately within each phase and refitted after excluding patients whose RT was never documented.

Giving imaging every chance. When imaging added nothing, we tested five explanations. *The label*: progressions documented on clinical grounds alone were censored, giving an imaging-defined label. *The segmentation*: we checked volumes, labels and dates against their sources and resegmented all 594 complete MU scans with the MONAI BraTS 2021 SwinUNETR and with BrainSegFounder. *The cohort*: models trained on all MU scans were applied unchanged to LUMIERE; without treatment dates, the transportable model used clinical variables, timing and volumes (Supplementary S13). *The modality*: in CFB-GBM, with one row per patient, we tested whether ADC radiomics add to conventional radiomics for volumetric RANO 2.0 progression between the two scans after RT (Supplementary S18). *The encoder*: we fine-tuned the BraTS 2021 SwinUNETR on

the 120-day label inside the shared folds. The two deepest stages and a small head were trained on a cube around the lesion and cavity, with fixed hyperparameters and early stopping on held-out training patients; four networks per outer fold gave out-of-fold scores for stacking onto the core model (100 fine-tunes per arm). A copy trained on shuffled outcomes showed what an uninformative score does to the stack (Supplementary S22).

The audit layer was evaluated on two prespecified quantities: conformal coverage on test scans with a known outcome (P1), and whether withholding the least confident 30% of scans raises the AUC of the rest more than removing the same number at random within each phase (P2). It also probed shortcuts by refitting without the indicators for unknown markers and with treatment or timing alone (Supplementary S15). Every fitted step, including the conformal calibration split, stays inside the shared training folds.

The reports were evaluated on 249 grade 4 records from a cross-validation repeat not used in development. The prompt and checker were developed on the 59 scans of other grades and then frozen. The primary quantities were the share of reports the checker passes (no unsupported claim and every active flag stated) and the share that contains every item of the required content listed above. We also ran PRISM's original prompt on the same records, a smaller model (Qwen3.5-9B), and 30 perturbed records to check that a report follows a changed risk or flag. No public set of surveillance reports exists, so we used 230 glioma MRI reports from RadGenome-Brain MRI and the MRI descriptions of 181 published glioblastoma case reports (MultiCaRe) for form only (Supplementary S21).

3. Results

3.1. Cohorts

Table 1 describes the cohorts. In the primary population, 249 grade 4 scans from 126 patients, 93 scans were followed by documented progression within 120 days. The scans spread across the phases of care: 76 were taken before RT, 6 during RT, 98 within 12 weeks after RT and 69 later.

3.2. One split overstated performance

Rescored with the final cohort and labels, the first split was generous to every model (Figure 3A; Table 2). PRISM's initial radiomics model reached 0.80, but the core model, which uses no imaging, reached 0.87, and age, sex, IDH and MGMT alone reached 0.79. The intervals were wide and overlapped.

The test set rewarded any model that recognises the phase of care. Of its 60 negative scans, 36 were taken before RT and none of them was followed by documented progression, whereas 10 of the 68 scans before RT with a known outcome in the whole cohort were (15%; Figure 3B). So 60% of the positive–negative pairs set a progression scan against a fresh postoperative scan, and a rule with no training at all, "the scan was taken after RT", scored 0.740 on this test set.

Across 500 splits of the same design (Figure 4A; Table 2), PRISM's initial radiomics pipeline, reselecting its features on each split, scored higher on the first split than on 92% of the others, and the core model ranked at the 91st percentile. Each split's AUC followed the mix of phases in its test set (Figure 4B): about two thirds of the first split's excess came from its mix of scans, and the rest from its particular patients. Under repeated cross-validation the same core model scored 0.720 in

all gliomas, against 0.871 on the first split.

3.3. The phase of care carries the signal

Under repeated cross-validation, discrimination in grade 4 rose from 0.585 with clinical features alone to 0.644 with timing and 0.698 with treatment status: the core model (Figure 5A; Table 2). It was calibrated in the large (ratio of observed to expected events 1.04; slope 0.89; Supplementary S12), and gradient boosting did not beat ridge.

Most of this signal is the phase of care. Four phase categories recovered most of the gain over clinical features (R2 0.664 against R0 0.585; Figure 6A), and treatment features alone scored 0.701, as high as the full core model. Within the surveillance cohort, where the phase varies little, no base model discriminated well (0.549 to 0.604).

The phase signal survives censoring, but lies between phases. Of the 39 censored grade 4 scans, 38 were taken before RT, from 31 patients whose RT and chemotherapy were never documented. The 37 patients in that group were older, were scanned around surgery and had a median follow-up of 36 days after each scan (317 days for patients with RT documented); 14 died without documented progression. Censoring weights estimated within each phase raised the gain of R2 over R0 from +0.080 to +0.124, and excluding these patients left +0.057, so the prespecified rule for a robust finding was met (Table 4). If every censored scan had progressed, the gain would vanish, which seems unlikely but cannot be excluded. And the phase separates scans between phases, not within them: counting only pairs within the same phase, R2 scored 0.577 against 0.595 for R0 (Supplementary S20).

In the intended use. On scans taken on or after the start of RT (173 grade 4 scans from 86 patients, 88 events, 1 censored), the corrected core model, trained on all scans, scored 0.601 (0.515–0.686): above chance, but modest. It was calibrated in the large (observed to expected 1.01) but spread its predictions too widely (slope 0.53), because part of their range encodes the difference between scans before and after RT; refitting within the population did not improve discrimination (0.595). Within phases it scored 0.631 in the 12 weeks after RT and 0.583 later, and 0.616 over pairs within the same phase, so little of its discrimination here comes from ranking one phase against another (Supplementary S23). The rule for scan-day eligibility removed 38 of the 39 censored scans and all 14 deaths without progression, so the censoring question largely disappears in the intended use. In the full cohort, phase-specific weights or counting deaths as non-events raised absolute AUCs (corrected model 0.734 and 0.706) but moved no imaging contrast by more than 0.011 (Table 4).

3.4. Imaging adds nothing measurable

No imaging branch showed a measurable gain (Table 3; Figure 5B). In the prespecified primary analysis, nested radiomics changed the grade 4 core model by −0.032, and in grade 4 none of the 70 configurations had a CI above 0. Frozen pretrained encoders changed it by −0.004. Their features did carry an image signal (0.652 alone, against 0.566 with untrained weights), but it tracked tumour volume ($r = 0.58$). Radiomics also made the model overconfident (calibration slope 0.60). A model can give imaging credit without needing it: where volumes, morphology and change were included they took about half of the SHAP credit, yet the models scored no better than the core model (Figure 5C).

None of the five explanations rescued imaging (Table 4).

- **After RT**, where the phase cannot do the work, adding all imaging measurements to clinical and phase variables moved the AUC from 0.549 to 0.610, within the range of the permutation null ($p = 0.21$; Figure 6B).
- **The label** was not the bottleneck: with progression on clinical grounds alone censored, the core model plus imaging scored 0.722 against 0.696.
- **Nor was the segmentation.** Our volumes reproduced the dataset's own table ($r \geq 0.996$), yet across all gliomas imaging separated the progression scan itself from verified negatives with an AUC of only 0.68, and ET was larger at the progression scan than at the scan before in only 49% of 74 pairs. Resegmentation agreed with the original masks (median Dice 0.80 for ET) but did not improve detection (0.665; Figure 7; Supplementary S14).
- **The external cohort** showed no imaging effect: on LUMIERE the locked model with volumes scored 0.608, against 0.610 for clinical variables and timing alone (Supplementary S13).
- **Diffusion** did not change the result: in 77 CFB-GBM patients with ADC (19 events), ADC radiomics added +0.007 to conventional radiomics.
- **Training the encoder on the task** did not help either. The fine-tuned image score alone reached 0.644, again tracking tumour volume ($r = 0.57$) more than the core model ($r = 0.13$), and added +0.009 to the core model, at most about +0.04 within its interval. The shuffled-outcome control alone scored 0.476, as intended, but its stack gained +0.005, slightly above the limit of 0 we had set for a valid run: an uninformative score buys about half of the observed gain. Other weights, all gliomas, and a network that also saw the previous scan gave −0.005 to +0.017 (Supplementary S22).

3.5. What the audit found

The audit layer reproduced the core model's predictions exactly, and the contributions summed to every logit. Conformal coverage met its target (P1; Table 4), but almost every scan before RT received the set {0}, so all 5 grade 4 progressions before RT fell outside their sets. Selective prediction did not help (P2 not met): withholding the least confident scans raised the AUC of the rest only by changing the phase mix, since it kept 97.6% of scans taken before RT (Figure 8B).

The audit's value was what it found, none of which showed in the cross-validated AUC.

(1) A missingness shortcut: IDH was unknown in none of 93 progression scans but in 6 of 117 scans without progression, and removing the indicators for unknown markers lowered the grade 4 AUC by 0.023.
(2) A treatment-course model: treatment features alone scored 0.701.
(3) A molecular look-ahead: IDH and MGMT were used at 8 grade 4 scans taken 1 to 58 days before diagnosis.
(4) The cause of external miscalibration: IDH was unknown in 40.9% of LUMIERE scans, and the same shortcut explained most of the gap between observed and expected events there.
(5) A documentation leak: the flag for the 12 weeks after RT was missing when the RT end date was missing, which told a scan whose RT would be documented later apart from one whose RT never would be.

Removing these gave the *corrected core model*, PRISM's reference configuration: 0.678 (0.596–0.755) in grade 4, −0.020 against the original, equally well calibrated (observed to expected 1.04, slope 0.85), and 0.706 in all gliomas. On LUMIERE it scored 0.635 against 0.608 and fixed the calibration

(observed to expected 0.99, was 1.33; slope 0.97, was 0.38).

3.6. Reports that can be checked

On the 249 evaluation records, the checker passed all 249 reports (1.000; Wilson 95% CI 0.985–1.000), none omitted any of the 262 active flags or contained forbidden content, and 247 contained every item of the required content (Table 5; Figure 8A). The two exceptions named a flag without its reason. Reports had a median of 190 words, about a minute to read, at a reading grade of 9.7. Grounding mattered more than the model: PRISM's original prompt, given the same facts, gave management advice in 99% of its reports, a diagnosis in 51% and a RANO category in 41%, while the smaller model with the grounded prompt passed 243 of 249. Reports followed the changed field in all 30 perturbed records. A first grounded prompt failed the checker in 13 of 249 reports for one reason, a risk sentence that stated the horizon without citing its field; the final prompt cites a single field holding the whole risk statement (Supplementary S19 and S21). No clinician has yet read the reports.

4. Discussion

PRISM was built to predict progression from surveillance MRI, and its evaluation changed what PRISM is: a modest clinical and treatment-course model whose every estimate comes with a record of what it knew and a report that can be checked against it. Three lessons emerged.

A single split can measure the split. The first 0.80 was real for that test set, but the test set was easy: 60% of its pairs set a progression scan against a fresh postoperative scan, a contrast that any model able to recognise the phase of care will win. The same split lifted the core model from 0.720 under repeated cross-validation to 0.871. This adds a mechanism to the known instability of single splits (An et al. 2021): when phases with very different event rates are mixed, the composition of a small test set explains much of the variance, and a split search that matches the number of scans and positives does not control it. We suggest four reporting rules for this task: repeated grouped cross-validation; the distribution over splits whenever a single split is reported; results within phases of care; and a clinical comparator that knows the treatment course.

The phase of care is a confounder. A scan before RT is rarely followed by documented progression within 120 days; a scan in the 12 weeks after RT often is, partly because of pseudoprogression and the follow-up it triggers (Brandes et al. 2008; Taal et al. 2008; Wen et al. 2023). Four phase categories recovered most of the core model's gain, and two published clinical comparators also found time since RT influential (Gomaa et al. 2025; Leone et al. 2023). Imaging models that pool scans from all phases can therefore report high AUCs for recognising the phase rather than tumour biology. The signal survived censoring corrections but ranks scans between phases, not within them, and part of it reflects who stays in follow-up. This is why PRISM places every scan in its phase, flags the 12 weeks after RT, and is intended for scans from the start of RT onwards. There its AUC is 0.60, which shows how much of the apparent discrimination on mixed cohorts comes from the phase of care.

Imaging, as tested here, showed no gain. Even at the scan where progression was documented, volumes separated it from negatives with an AUC of only 0.68. Three features of the data explain this. The label is a documentation event: one in eight grade 4 events was documented on

clinical grounds alone. Volume series were noisy: ET changed more between scans less than 14 days apart (median 3.2-fold) than between scans 45 to 90 days apart (Figure 7C–D). And the cohort is small: by the Riley criteria even the core model would need about ten times more scans (Riley et al. 2020). Small effects cannot be excluded: the upper confidence bounds for imaging reach 0.02 to 0.16, and training the encoder on the label gave +0.009. Where imaging performs well in the literature, it uses diffusion and perfusion to classify a lesion that is already suspicious (Kim et al. 2019; Taylor et al. 2022), a different question from forward risk at every scan. PRISM keeps its imaging branches as optional modules so that larger cohorts can test them again. Attribution is no guide: imaging took half of the SHAP credit without improving discrimination, so a claim of imaging value needs a paired difference with an interval.

What PRISM offers. At an AUC of about 0.70, PRISM does not offer better triage: abstention raised the AUC of the remaining scans only by changing their phase mix, and the conformal sets met their target mainly by calling every scan before RT negative. What it offers is a record of what the model knew on the scan day, where the scan sits in the treatment course, how uncertain the estimate is and which inputs drove it. Running the audit found a missingness shortcut, a molecular look-ahead, a documentation leak and the cause of the external miscalibration, none of which the AUC revealed. Grounding the language model in that record and checking every sentence against it gave reports that restate the record without verdicts or advice, and that put the number in context: how it compares with the average, how it changed since the last scan, what the model could not see and why the phase matters. We think this is the honest use of a modest model: to make its reasoning and its failure modes visible, not to claim better decisions. Whether clinicians find the reports useful is the next question; a protocol for a randomised reader study is in Supplementary S21.

Comparison with other work. The grade 4 core model (0.698) is close to clinical comparators for neighbouring tasks (0.69 in Leone et al. 2023; 0.727 externally in Gomaa et al. 2025). It is below the single test set result of 0.80 on MU-Glioma-Post (Christodoulou et al. 2026), which used a different label (the progression status of each scan) and which we did not re-evaluate. Higher AUCs for forward recurrence prediction elsewhere use different cohorts, horizons and units, and some fall externally (0.887 to 0.762 in Chen et al. 2026).

Limitations

The development cohort comes from one centre, with 126 grade 4 patients and 93 events at 120 days, far below recommended sample sizes. The outcome is documented progression from routine records, not a blinded RANO read, and its timing depends on scheduling. MU images carry no acquisition metadata, so scanner effects could not be studied. Censoring was concentrated in patients whose RT was never documented; phase-specific weights and their exclusion left the phase finding intact, but censoring that depends on the outcome itself cannot be ruled out. Our bootstrap intervals resample fixed held-out predictions and do not include the variability of refitting, so they are likely too narrow. The external cohort is small, uses a different segmenter and lacks treatment dates, so the full core model was not tested externally. Several imaging tests were added after the first imaging result was known, although each had its rule fixed before its own run. Radiomics used a fixed bin count rather than the fixed bin width recommended by IBSI (Zwanenburg et al. 2020). The fine-tuned encoder was trained on about 100 scans per network with fixed hyperparameters,

and its shuffled-outcome control slightly missed its validity limit. The ablation and phase analyses use the original core model; the corrected model scores 0.020 lower. The report checker is lexical: it verifies numbers, citations, flags and forbidden patterns, not meaning. The required content was defined by us, so meeting it shows that the reports say what we specified, not that they help; the public reference reports describe preoperative or published cases rather than routine surveillance; and no clinician has assessed the reports. Finally, at an AUC of 0.60 in its intended use, PRISM is not ready to guide care; a triage model would need larger multicentre cohorts, a label read from images by experts, and perfusion or diffusion imaging, and would need to be tested prospectively.

Conclusion

PRISM's first AUC of 0.80 came from one favourable split. Under repeated cross-validation its core of clinical and treatment-course variables reached about 0.70, or 0.68 after removing leaks found by its own audit, and radiomics, deep features (frozen or fine-tuned), resegmentation, a cleaner label and diffusion added no measurable discrimination. The phase of care drives discrimination on this task and should be reported as a stratum and treated as a confounder. In surveillance from the start of RT, PRISM's intended use, its corrected core model reached an AUC of 0.60, which is not enough to guide care. What PRISM provides is a way to build and test such models honestly: every prediction is recorded, explained and reported in a form that can be checked, and the audit exposed leaks that the AUC hid.

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Disclosure statement

[PENDING AUTHOR DECISION: competing interests, or "The authors report there are no competing interests to declare."] If any author serves on the editorial or advisory board of this journal, state it here.

Declaration of generative AI use

[DRAFT, PENDING AUTHOR CONFIRMATION:] Generative AI is part of the system studied. PRISM's report generator uses a local language model (Qwen3.8-27B; Qwen3.5-9B in a comparison arm), and a second local model (Gemma 4 31B) rated reports in an exploratory check (Supplementary S21); all ran offline on the authors' hardware, saw only derived audit records, and are described in Sections 2.1 and 3.6. An AI assistant (Claude, Anthropic) was used to write and run analysis code, to read 15 generated reports against their records (Supplementary S19), and to draft and edit the manuscript under the authors' direction. Patient data did not leave the authors' computing cluster. The authors reviewed all code, results and text and take full responsibility for the content of this article.

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Ethics statement

[PENDING AUTHOR DECISION: secondary analysis of public de-identified datasets (MU-Glioma-Post, LUMIERE, CFB-GBM); state the institutional determination or waiver and cite the source datasets’ approvals.] No registered protocol. Analysis plans and dated plan snapshots are in the repository (Supplementary S1). Patient and public involvement: none.

Data availability statement

[PENDING AUTHOR DECISION: MU-Glioma-Post (TCIA, doi:10.7937/7K9K-3C83), LUMIERE (figshare; Suter et al. 2022) and CFB-GBM (TCIA, doi:10.7937/v9pn-2f72) are public; derived tables, fold assignments and locked model files; state what is released and give a DOI or other persistent identifier.] Code: [PENDING: public repository link and release tag.]

Supplemental material

Supplemental data for this article (Supplementary S1–S23, Tables S1–S45) are provided as a separate file.

Notes on contributors

[PENDING: a short biographical note for each author (no more than 200 words each).]

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Tables

Table 1. Cohort characteristics. Index scans are MRIs before first documented progression. Values are n (%) or median [IQR]. Patient rows use each patient's first index scan. Full table with definitions: Supplementary S2.

	MU all gliomas	MU grade 4	LUMIERE
Patients / index scans	155 / 308	126 / 249	61 / 132
Index scans per patient, median (range)	2 (1–6)	2 (1–6)	2 (1–10)
Age, years	61 [47–69]	62 [53–70]	62 [53–70]
Male sex	80 (52%)	66 (52%)	32 (52%)
IDH wild type / mutant / unknown	120 (77%) / 22 (14%) / 13 (8%)	112 (89%) / 8 (6%) / 6 (5%)	38 (62%) / 1 (2%) / 22 (36%)
MGMT methylated / unmethylated / unknown	53 (34%) / 73 (47%) / 29 (19%)	45 (36%) / 62 (49%) / 19 (15%)	26 (43%) / 28 (46%) / 7 (11%)
WHO grade 4	126 (81%)	126 (100%)	61 (100%)
Radiotherapy documented	102 (66%)	89 (71%)	not recorded
Initial chemotherapy documented	96 (62%)	88 (70%)	not recorded
Days from origin to scan[a]	97 [46–186]	96 [54–182]	119 [82–226]
RT completed by scan day (scans)	185 (60%)	167 (67%)	not recorded
Within 12 weeks after RT[b] (scans)	104 (34%); unknown 94[c]	98 (39%); unknown 65[c]	27 (20%)
Gap to next MRI, days	77 [50–103]	74 [49–98]	91 [77–117]
Death without documented progression (scans)	22 (7%)	20 (8%)	7 (5%)
120 d: progression / no progression / censored	100 / 156 / 52	93 / 117 / 39	47 / 76 / 9
90 d	65 / 195 / 48	62 / 152 / 35	24 / 101 / 7
180 d	144 / 105 / 59	131 / 75 / 43	62 / 61 / 9

[a] MU: days since diagnosis; LUMIERE: days since first surgery.
[b] MU: 0–84 days after the documented RT end date; LUMIERE: rater's "< 3 months after RT" flag.
[c] RT never documented (92 scans, all gliomas; 63, grade 4) or RT end date missing (2 scans). Every scan is still assigned a phase of care (Section 2.3).

Table 2. The same models on a single split, across 500 splits and under repeated cross-validation. IPCW AUC(120 d) with 95% bootstrap over patients CI unless stated. Single split = PRISM's first evaluation split (all gliomas; 29 test patients, 82 scans). 500 splits = PRISM v1's split design (naive label; for PRISM's initial radiomics model, its pipeline with features reselected per split). Core model = PRISM's reference model of clinical, timing and treatment variables (the reference for every imaging comparison). 5 × 5 CV = repeated cross-validation grouped by patient on the final cohort.

Model	Single split: naive AUC	Single split: IPCW AUC [95% CI]	Δ vs core ridge [95% CI]	500 splits: naive mean ± SD (2.5–97.5%)	5 × 5 CV, all gliomas	5 × 5 CV, grade 4
PRISM initial radiomics model	0.802	0.817 [0.690, 0.920]	−0.034 [−0.186, +0.109]	0.720 ± 0.055 (0.610–0.819)	n/a (fixed model)	n/a (fixed model)
Core model (ridge)	0.871	0.851 [0.735, 0.947]	–	0.801 ± 0.052 (0.694–0.895)	0.720 [0.638, 0.792]	0.698 [0.615, 0.774]
Core model (GBM)	0.844	0.824 [0.690, 0.916]	−0.027 [−0.127, +0.062]	not run	0.718 [0.642, 0.792]	0.687 [0.601, 0.768]
Core + legacy radiomics	0.862	0.845 [0.746, 0.936]	−0.006 [−0.081, +0.090]	0.799 ± 0.053 (0.674–0.892)	0.743 [0.678, 0.803]	0.703 [0.627, 0.774]
Clinical only (ridge)	0.791	0.782 [0.638, 0.897]	−0.069 [−0.215, +0.058]	not run	0.619 [0.540, 0.696]	0.585 [0.496, 0.666]

Table 3. Do PRISM's imaging branches add to its core model? Paired ΔAUC with 95% bootstrap over patients CI. Reference: PRISM's core model (clinical + timing + treatment, ridge) unless stated in the section heading. Grade 4 (G4) is primary. * = CI excludes 0 (uncorrected; 3 of the 32 intervals in Figure 5B). The fine-tuned encoder and CFB-GBM rows are not in Figure 5B. Decision rules and answers: Table 4.

Analysis	G4 ΔAUC [95% CI]	All gliomas ΔAUC [95% CI]
Radiomics branch vs core model		
Nested over 70 configurations (prespecified primary)	−0.032 [−0.084, +0.017]	+0.028 [−0.017, +0.071]
Legacy top-10, concatenated (single configuration)	+0.005 [−0.059, +0.067]	+0.024 [−0.026, +0.075]
Legacy PCA-10, stacked (best post hoc, optimistic)	+0.023 [−0.013, +0.057]	+0.035 [+0.002, +0.068]*
Frozen deep encoder branch vs core model		
Nested over 32 (BraTS21 SwinUNETR + BrainSegFounder)	−0.004 [−0.069, +0.063]	+0.011 [−0.032, +0.056]
BraTS21 SwinUNETR, lesion PCA-10 stacked	+0.016 [−0.029, +0.059]	+0.017 [−0.014, +0.050]
BrainSegFounder, lesion PCA-10 stacked	+0.014 [−0.029, +0.057]	+0.024 [−0.013, +0.064]
Untrained encoder, nested (control)	−0.049 [−0.085, −0.014]*	−0.009 [−0.032, +0.009]
Fine-tuned deep encoder branch vs core model		
BraTS21 SwinUNETR, fine-tuned (prespecified primary)	+0.009 [−0.022, +0.040]	+0.017 [−0.009, +0.044]
BrainSegFounder, fine-tuned	−0.005 [−0.030, +0.019]	–
BraTS21 SwinUNETR, fine-tuned with the previous scan	+0.014 [−0.010, +0.040]	–
BraTS21 SwinUNETR, shuffled outcomes (control)	+0.005 [−0.006, +0.015]	–
Imaging-defined label (progression on clinical grounds alone censored)		
Core + volumes/change vs core (prespecified rule)	+0.027 [−0.033, +0.082]	+0.008 [−0.036, +0.049]
Surveillance scans (after RT) only		
Clinical + phase + imaging vs clinical + phase (prespecified primary)	+0.061 [−0.038, +0.156]	+0.033 [−0.054, +0.128]
Core + imaging measurements vs core	+0.017 [−0.069, +0.093]	+0.005 [−0.075, +0.082]
Core + radiomics (nested) vs core	+0.005 [−0.070, +0.075]	+0.024 [−0.030, +0.076]
Resegmentation (reference: same model on MU masks)		
Clinical + phase + imaging, BraTS21 SwinUNETR masks	+0.030 [−0.008, +0.069]	+0.029 [−0.010, +0.069]
Clinical + phase + imaging, BrainSegFounder masks	+0.008 [−0.023, +0.040]	+0.003 [−0.025, +0.032]
LUMIERE, locked (reference: clinical + timing)		
+ volumes	−0.001 [−0.073, +0.057]	−0.006 [−0.075, +0.055]
+ deep features	+0.057 [−0.132, +0.235]	+0.064 [−0.011, +0.144]
+ volumes + deep vs + volumes	+0.089 [−0.026, +0.215]	+0.050 [+0.002, +0.109]*
CFB-GBM (77 patients, 19 events; reference: conventional radiomics)		
+ ADC radiomics (prespecified primary)	+0.007 [−0.019, +0.035] (permutation p = 0.151)	–

Table 4. Prespecified questions, rules and answers. Each rule was fixed before its analysis was run (Supplementary S1). Grade 4 glioma unless stated. ΔAUC is the paired difference in IPCW AUC at 120 days with 95% bootstrap CI over patients; "CI > 0" means the interval excludes 0. Core = PRISM's core model; R0 = clinical; R2 = clinical + phase of care; I2 = all imaging measurements except radiomics.

Question	Rule	Result	Answer
Does imaging add to PRISM's core model?			
Radiomics (primary development analysis)	Core + nested radiomics vs core; report the paired ΔAUC	−0.032 [−0.084, +0.017]	No gain
Imaging after RT, where the phase cannot help	R2 + I2 vs R2 in surveillance scans; permutation test	+0.061 [−0.038, +0.156]; p = 0.21	Not supported
Is the label the bottleneck?	Imaging-defined label: imaging gains ≥ 0.03 with CI > 0	+0.027 [−0.033, +0.082]	Not met
Is the segmentation the bottleneck? (all gliomas)	New masks raise detection of the progression scan by ≥ 0.05 with CI > 0, or ET is larger at progression in > 65% of pairs	Detection 0.681 to 0.665; ET larger in 45%	Not met
Does diffusion add? (CFB-GBM)	ADC radiomics adds ≥ 0.03 to conventional radiomics with CI > 0	+0.007 [−0.019, +0.035]; p = 0.151	Not met
Does fine-tuning the encoder help?	Core + fine-tuned score gains ≥ 0.03 with CI > 0; validity: shuffled-outcome control scores about 0.5 alone (CI covering 0.5) and gains ≤ 0 in the stack	+0.009 [−0.022, +0.040]; control 0.476 [0.438, 0.514] alone, +0.005 [−0.006, +0.015] in the stack	Not met; control limit missed
Do the findings survive censoring?			
Phase signal under censoring	R2 − R0 ≥ 0.040 with phase-specific censoring weights and after excluding patients whose RT was never documented	+0.124 [+0.039, +0.232]; +0.057 [+0.009, +0.117]	Robust
Imaging contrasts under censoring sensitivities	No imaging contrast gains ≥ 0.03 with CI > 0 with phase-specific weights or with deaths counted as non-events	Largest change in any contrast 0.011; radiomics −0.021 and −0.035; fine-tuned +0.011 and +0.009	Conclusions hold
Does the audit layer work?			
P1: conformal coverage	Coverage ≥ 0.78 on test scans with a known outcome	0.841 [0.790, 0.885]	Met
P2: selective prediction	Abstaining on the least confident 30% beats random removal within phase, CI > 0	+0.006 [−0.023, +0.036]	Not met
Are the reports faithful and complete?			
Faithful and complete	Checker passes ≥ 95% of reports; every required content item in ≥ 90%; more required content than the first grounded prompt (paired CI > 0); forbidden content ≤ 2% and no active flag omitted[a]	100%; 99.2%; +0.325 [+0.312, +0.337]; 0%, none	Met

[a] Evaluated on 249 records from a cross-validation repeat not used in development. A first grounded prompt, evaluated on another repeat against a checker pass target of ≥ 0.95, passed 236/249 = 0.948 [0.913, 0.969] and was revised (Section 3.6).

Table 5. Evaluation of PRISM's reports on 249 grade 4 records from a cross-validation repeat not used in development. Required content was fixed before the evaluation; the checker is deterministic.

	Reports
Passed the checker (no unsupported claim, every active flag stated)	249 (100%)
Contained every applicable required item	247 (99.2%)
Required content: reports containing each item	
Risk with its 95% interval and the 120-day horizon	249 of 249
Reference frame (risk tier or the model's average risk)	249 of 249
Phase of care, and why it matters after RT	249 of 249
Main drivers of the risk	249 of 249
What the model did not see	249 of 249
Each active flag with its reason	178 of 180
Change since the previous scan	123 of 123
Input not known on the scan day	8 of 8
Safety and form	
Forbidden content (diagnosis, RANO category, advice)	0
Active flags omitted	0 of 262
Words, median [IQR]	190 [158–214]
Reading grade (Flesch–Kincaid), median	9.7
Reading time at 200 words per minute, median	57 s

Figure legends

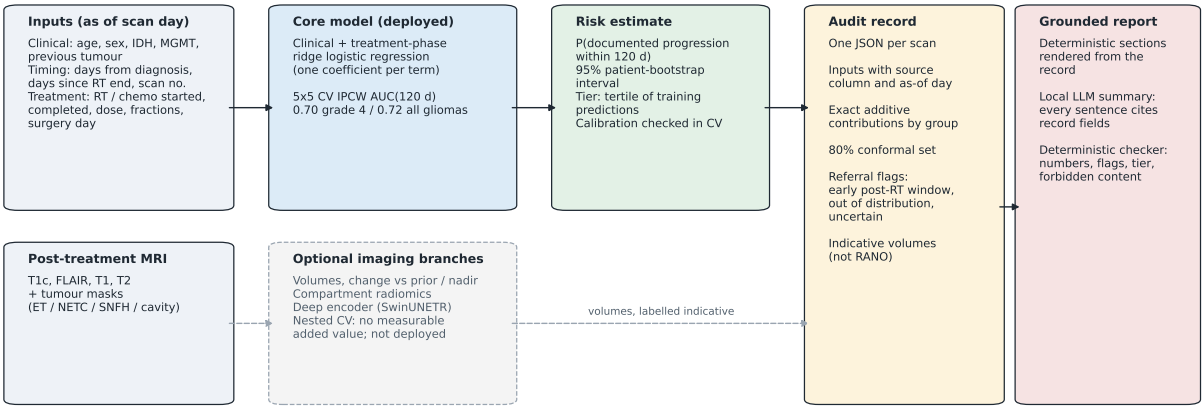


Figure 1. PRISM architecture and data flow. Inputs read as of the scan day feed the reference core model (ridge on clinical and treatment-course variables). The imaging branches (volumes, radiomics, deep encoder; dashed) are optional and switched off in the reference configuration; volumes enter the record only as indicative values. The risk, with its interval and tier, goes into the per-scan audit record (contributions, 80% conformal set, referral flags), from which the grounded report is written.

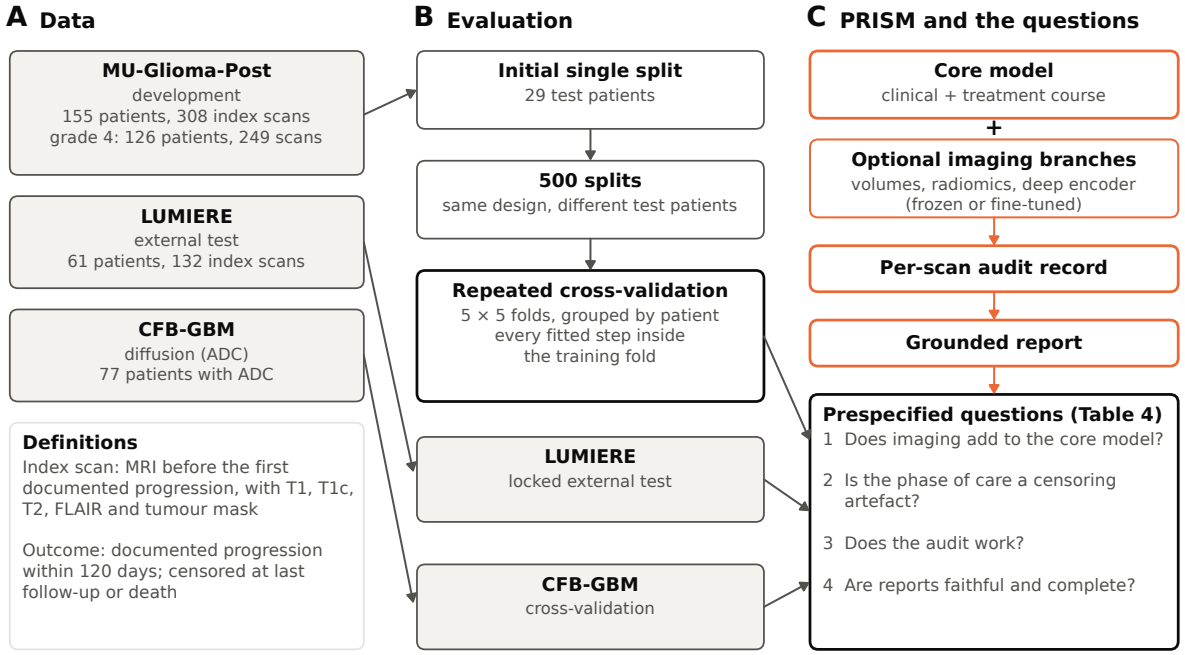


Figure 2. Study design. (A) MU-Glioma-Post was used for development, LUMIERE as a locked external test and CFB-GBM for the diffusion analysis. (B) Evaluation moved from one split to 500 splits of the same design and then to repeated cross-validation grouped by patient, with every fitted step inside the training fold. (C) PRISM's core model, its optional imaging branches, the per-scan audit record and the grounded report, and the questions tested against prespecified rules (Table 4).

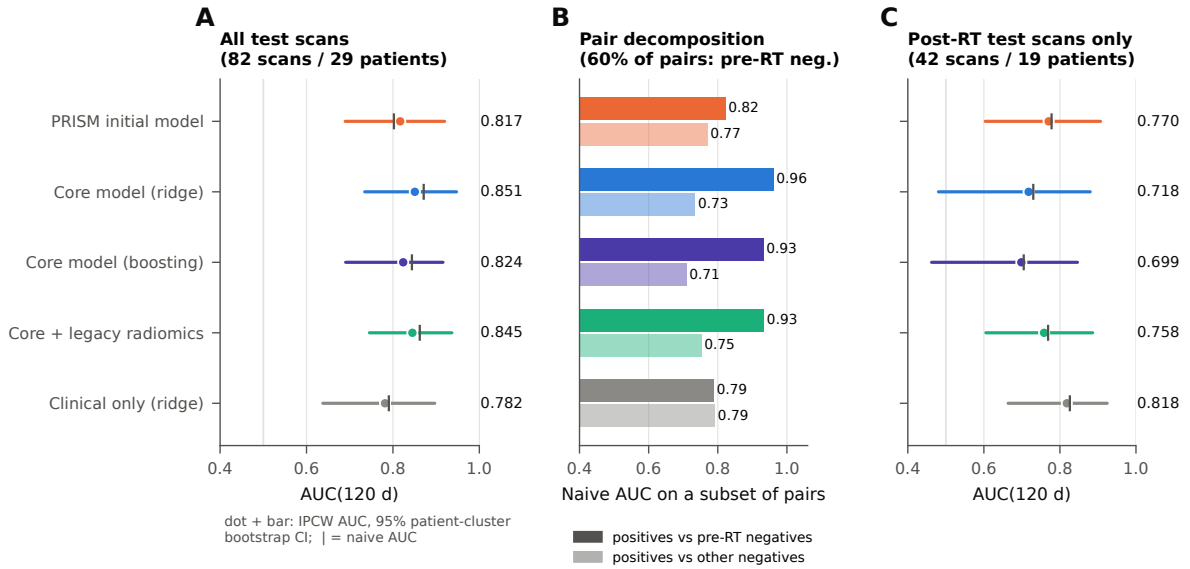


Figure 3. PRISM's first evaluation split. Discrimination on the first evaluation split (all gliomas; 126 training / 29 test patients, 82 test scans, 22 naive positives). Every model except PRISM's initial radiomics model was fitted once on the training patients; the initial radiomics model enters with its shipped test-set probabilities. (A) IPCW AUC with 95% bootstrap over patients CI (1,000 draws; dot and bar) and naive AUC (tick). (B) Naive AUC decomposed by the type of negative scan: 36 of the 60 negative test scans were taken before RT started, and none of them progressed, so 60% of positive–negative pairs compare a progression scan with a scan taken before RT; solid bars, positives against negatives taken before RT; light bars, positives against other negatives. (C) The same models on the 42 test scans taken after RT (19 patients). Intervals are wide and overlap throughout.

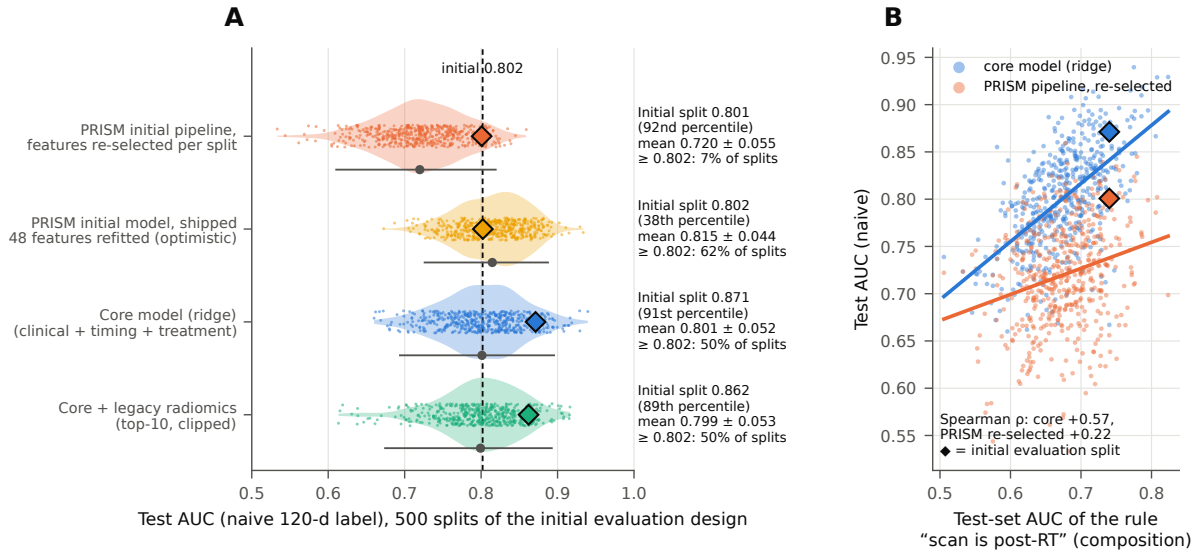


Figure 4. Where the first split sits among 500 splits of the same design. PRISM's split search was run for seeds 0–499; seed 62 reproduces the first split. (A) Test AUC (naive label) per split for PRISM's initial radiomics pipeline with features reselected per split (honest estimate), the shipped features refitted (optimistic upper bound), the core ridge model and the core model + legacy radiomics. Violin and points, 500 splits; grey dot and line, mean and 2.5–97.5% range; diamond, each model's own value on the first split with its percentile rank; dashed line, the reported 0.802. (B) Per-split AUC against the AUC of the no-training rule "scan is after RT" on that test set (higher = more negatives and fewer events before RT); lines, least-squares fits; Spearman ρ +0.57 (core model) and +0.22 (initial radiomics pipeline).

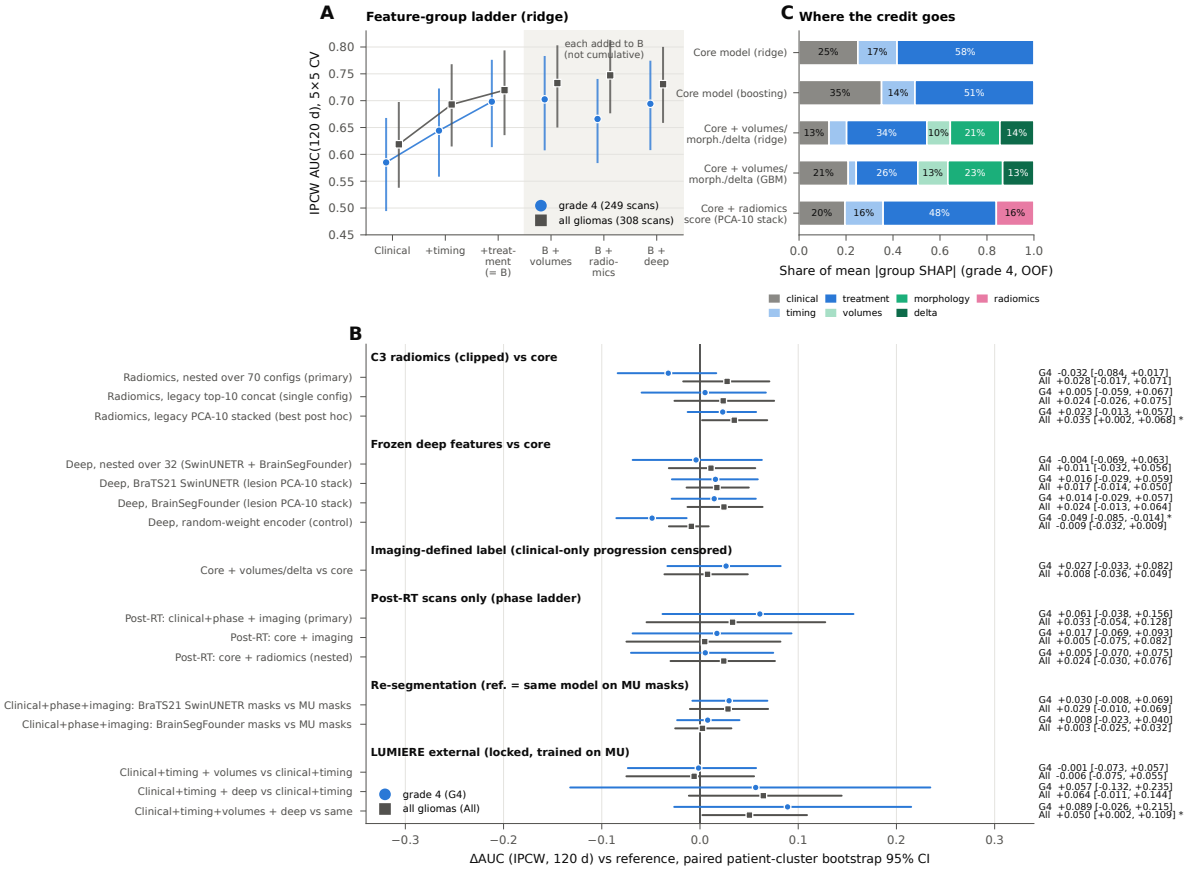


Figure 5. Component ablation: clinical, treatment-phase and imaging branches under repeated cross-validation. (A) Feature ladder (ridge; 5 × 5 repeated CV grouped by patient; IPCW AUC(120 d) with 95% bootstrap over patients CI): clinical → + timing → + treatment status on the scan day (= PRISM's core model, B); the last three points each add one imaging branch to B (volumes; radiomics, nested over 70 configurations; frozen deep features, nested over 32). (B) Paired ΔAUC with 95% CI against the reference named in each section (Table 3). Asterisks mark CIs excluding 0 (3 of 32, uncorrected; about 1.6 expected by chance). (C) Share of mean absolute group SHAP in grade 4 held-out explanations: imaging blocks take 45–51% of the credit in the full models without improving discrimination.

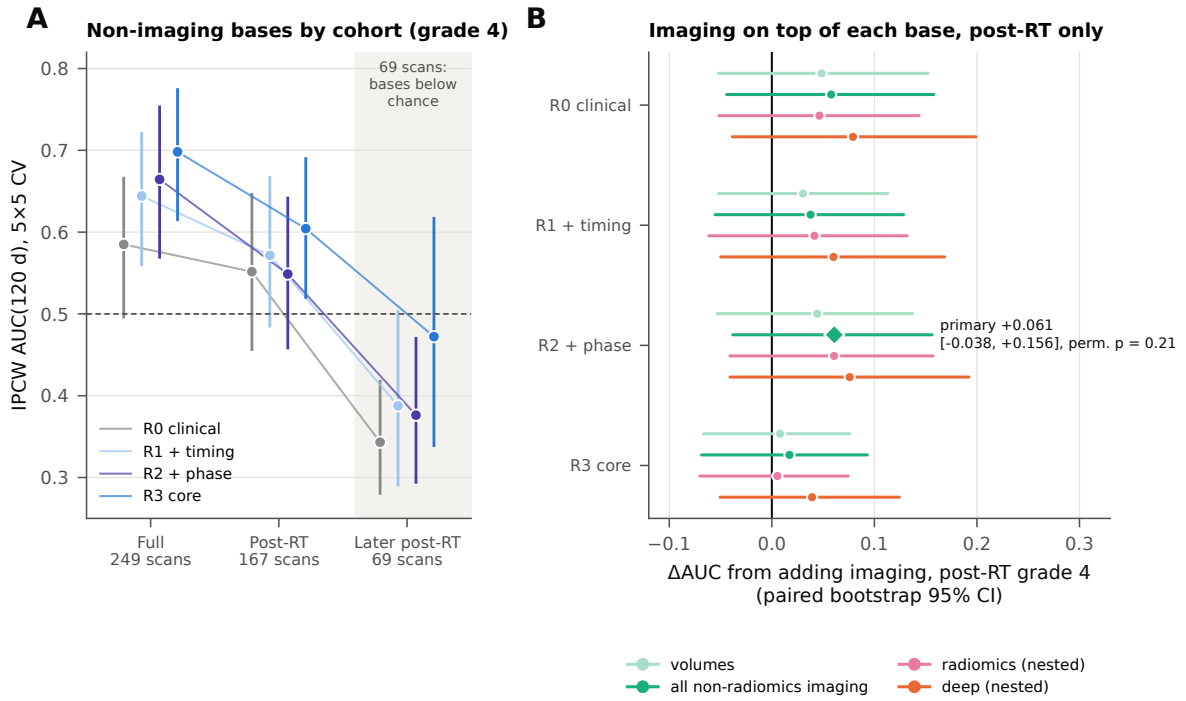


Figure 6. Phase of care. (A) Bases without imaging (ridge, grade 4) in the full cohort (249 scans), the surveillance cohort (scans after RT; 167 scans, 84 patients, 83 events) and the later surveillance cohort (> 12 weeks after RT end; 69 scans), each trained and evaluated within the cohort. R0 clinical; R1 + timing; R2 + four-level phase of care; R3 core model. In the later surveillance cohort the bases fall below 0.5, a known bias of cross-validation for nearly null models in small samples, so contrasts there are not interpretable. (B) ΔAUC from adding each imaging block (I1 volumes; I2 all imaging measurements other than radiomics; I3 radiomics; I4 deep) to each base within grade 4 surveillance scans. Diamond, prespecified primary contrast (R2 + I2 vs R2; Table 4); the permutation null itself averages +0.03.

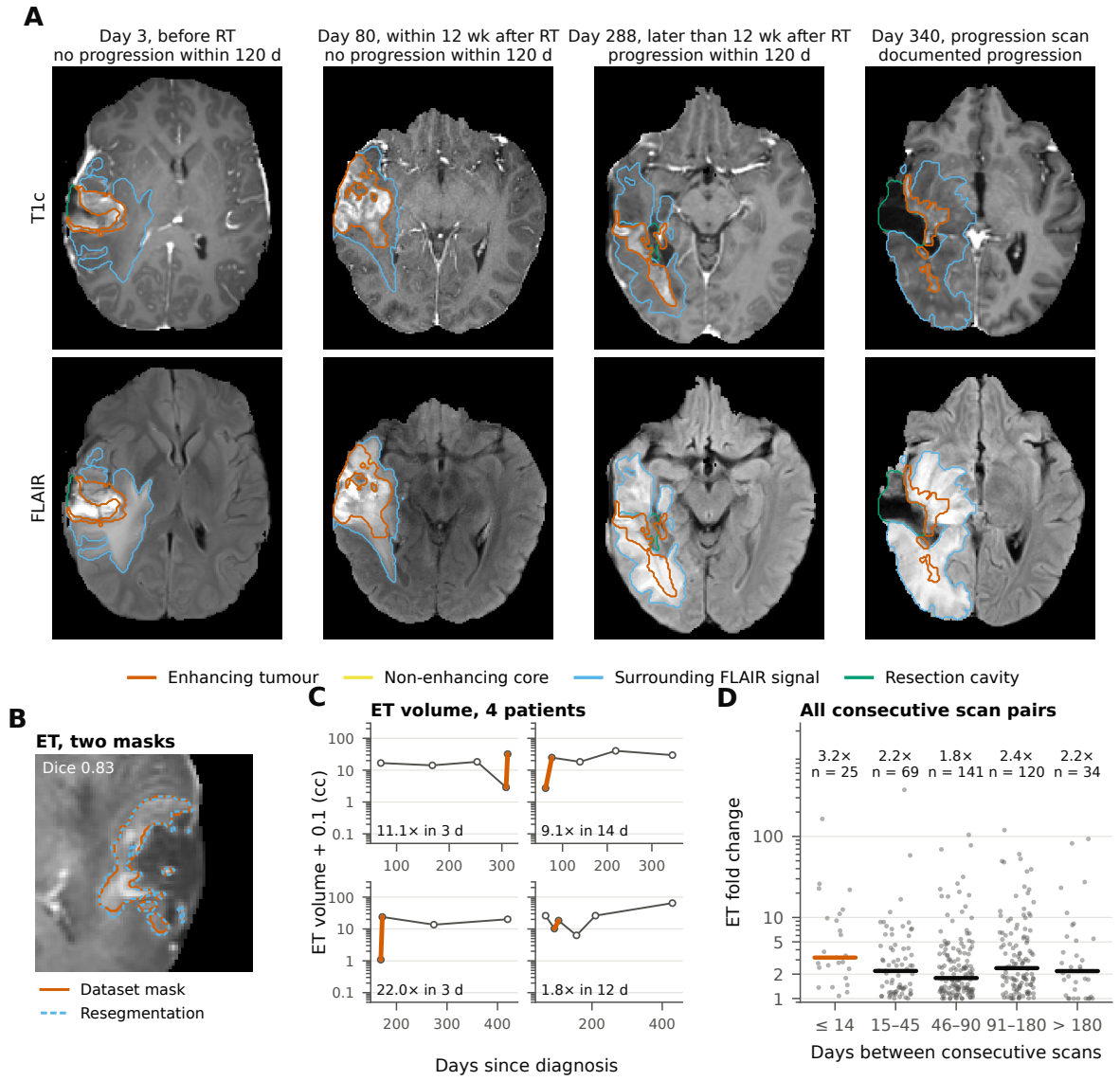


Figure 7. Why imaging struggles on this task. MU-Glioma-Post images (TCIA, CC BY 4.0; Mahmoud et al. 2025). (A) One grade 4 patient across phases of care, chosen by a fixed rule (the patient with a progression scan whose index scans cover the most phases): axial T1c and FLAIR at the slice of largest enhancing tumour, with mask contours, the day since diagnosis, the phase and the 120-day outcome. (B) Enhancing tumour from the dataset mask and from the BraTS 2021 SwinUNETR resegmentation, for the grade 4 scan with the median Dice (0.83). (C) Enhancing tumour volume over time in four patients with scans 14 days or less apart (highlighted, with the fold change). (D) Fold change in enhancing tumour volume between consecutive scans by interval, with medians: volumes changed by a median of 3.2-fold over 14 days or less, no less than over longer intervals.

A

PRISM surveillance report

PatientID_0003_Timepoint_1 | scan day 90 from diagnosis | early post-RT (0-84 days after the end of radiotherapy)

Risk of documented progression within 120 days

0.46 (95% interval 0.26-0.71)

Tier: middle third of development-cohort predictions (cut points 0.26 / 0.50). 80% conformal set: both outcomes (progression and no progression).

Referral flags

[!] early post-RT window: scan 25 days after the end of RT, inside the 84-day early post-RT window where treatment effects can mimic progression

[!] out of distribution: MGMT status unknown

[!] uncertain: the 80% conformal prediction set contains both outcomes

Action: flagged for clinician review

Contributions (exact additive log-odds; positive = towards higher risk)

Intercept -0.39 clinical +0.14 timing -0.34 treatment +0.43 = logit -0.17. Imaging: not part of this model.

-0.27 MGMT status unknown: yes (clinical)
-0.27 scan number = 1 (timing)
+0.16 RT started: yes (treatment)
+0.16 male sex: no (clinical)
+0.15 previous brain tumour: no (treatment)

Inputs as of the scan day (source: MU clinical workbook)

Age 57, female; IDH wildtype; MGMT unknown; RT started yes, completed yes; days since RT end 25; chemotherapy started yes. All 23 inputs known by the scan day: yes (each with source column and as-of day in the record).

Volumetric change (indicative, not RANO)

Enhancing 33.8 cc; FLAIR hyperintensity 44.3 cc.

First scan: no prior or nadir.

Summary (local LLM; every sentence cites record fields; checked deterministically)

FINDINGS: This scan is 90 days from diagnosis, 25 days after radiotherapy ended, in the early post-RT phase. In this window, treatment-related change can look like progression on MRI, which the model cannot separate. The model estimates a 46% (95% interval 26-71%) risk of documented progression within 120 days. This is in the middle third of the model's estimates on its training scans, above the average. The largest contributions are MGMT status unknown and scan number one, both towards lower risk. The model did not use the MRI images or tumour volumes, steroid dose, or symptoms. The scan is flagged for clinician review due to the early post-RT window, out of distribution, and uncertain status.

IMPRESSION: The model estimates a 46% risk, in the middle third above the average, with active flags for clinician review.

Faithfulness check: 8/8 claims supported; active flags mentioned: all; forbidden content: none.

Limitations

Model discrimination in development cross-validation AUC 0.70 (single institution, MU-Glioma-Post); the risk is the probability of documented progression within 120 days, not a diagnosis of progression or pseudoprogression; volumes are indicative, not a RANO assessment.

B

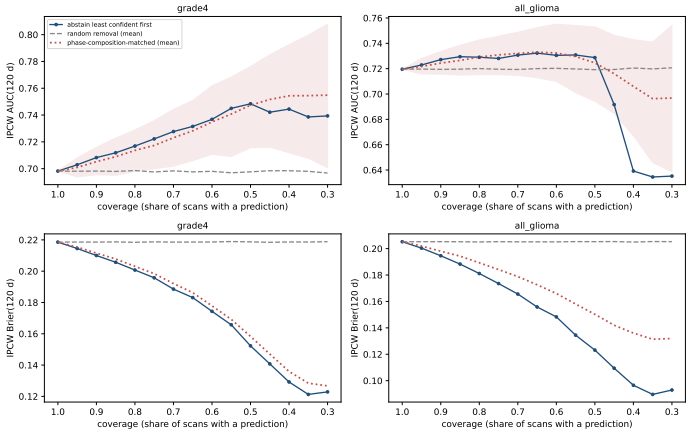


Figure 8. A grounded report and selective prediction. (A) Grounded report for the first evaluation scan in the 12 weeks after RT, chosen by a prespecified rule: sections rendered from the audit record and a local language model summary, with findings and an impression, in which every sentence cites record fields. The checker found every claim supported. By the required-content rule this is one of the 2 of 249 incomplete reports: it names the active flags without their reasons. (B) Risk–coverage curves for the core model (5 × 5 CV, pooled held-out predictions, mean over repeats), grade 4 (left) and all gliomas (right). Top, IPCW AUC(120 d); bottom, IPCW Brier score of the scans kept when the least confident scans (lowest max(p, 1 − p)) are withheld first (solid), against random removal (dashed) and random removal matched by phase (dotted; band, 2.5–97.5% of the matched draws). The gain over random removal is almost entirely matched by the phase-matched reference (prespecified P2 at about 70% coverage, grade 4; Table 4).